

Multivariable Calculus

UChicago Political Science Math Camp

Ryan Gibbons & Luke Cain

2025

Multivariable Calculus

Really, it's not that bad

- ▶ this is very true for derivatives
- ▶ this is kind of true for integrals

Functions

Recall our definition of a **function**:

A function is a mapping of elements in some set to elements in another set.

So far, we have been working with functions of the form:

$$f : \mathbb{R} \rightarrow \mathbb{R}$$

which we simplify into $f : x \rightarrow f(x)$ or $f : x \rightarrow y$.

Multivariable Functions

Multivariable functions¹ return an output given a series of inputs. More generally, we examine:

$$f : \mathbb{R}^n \rightarrow \mathbb{R}$$

Such that we “blend” n input numbers into a single output
 $f(x_1, x_2, \dots, x_n)$

This is not visually interpretable beyond $n = 2$.

¹to the extent that we are concerned about them

Example

We can represent the following as multivariate functions:

- ▶ The **wind chill** W_c is a function of temperature t and wind speed s
- ▶ Your **commute time** (from Pick, walking) C is a function of your numbered street n , cross street s , and walking speed w
- ▶ Your **grocery store bill** G is a function of price p_i and quantity q_i for each product $i \in \{\text{shopping list}\}$

Recap: Hints for Multivariable Functions

- ▶ Hint #1: use $\mathbb{R}^n$ for n input variables
- ▶ Hint #2: we are often just mapping something onto $\mathbb{R}$, which means that for the input(s), *we just recover a single numeric output.*
- ▶ Hint #3: don't try to visualize functions involving beyond $\mathbb{R}^3$ (and really not beyond $\mathbb{R}^2$)

Multivariable Calculus

calls for two expansions of our existing toolbox:

1. Partial derivation
2. Iterated integration

Both will take a common approach: *treat non-w.r.t. variables as constants!*

Partial Derivatives: Step 1



Partial Derivatives: Step 2



Partial Derivatives

What happens to the gauge when you move one (1) lever?

This is contingent on two primary things:

1. What direction you move that lever
2. Where all the other levers are set

These conditions are re-evaluated at every point!

Notation

We use the ∂ symbol to represent a partial derivative:

$$\frac{\partial f}{\partial a}$$

represents the **instantaneous rate of change** of f , given a change in *only* a . To take a second derivative, we use

$$\frac{\partial^2 f}{\partial a^2}$$

Why can't we use f' as notation?

How We Do It

To take a partial derivative, we follow simple steps:

1. Identify the variable of derivation:

$$\frac{\partial f}{\partial a} \left[2a^2 + a^3b \right] = \text{partial of } f \text{ w.r.t } a$$

2. Treat all other variables (i.e., b) as constants, then follow normal derivation rules for a :

How We Do It

To take a partial derivative, we follow simple steps:

1. Identify the variable of derivation:

$$\frac{\partial f}{\partial a} \left[2a^2 + a^3b \right] = \text{partial of } f \text{ w.r.t } a$$

2. Treat all other variables (i.e., b) as constants, then follow normal derivation rules for a :

$$\begin{aligned} &\implies 2 \cdot a^{2-1} + b \cdot 3 \cdot a^2 \\ &\implies \frac{\partial f}{\partial a} = 4a + 3ba^2 = a(4 + 3ba) \end{aligned}$$

Recall that the derivative of a constant is 0, so $\frac{\partial f}{\partial a}$ for a term excluding a is 0.

Some Evaluation

We've recovered

$$\frac{\partial}{\partial a} \left[2a^2 + a^3b \right] = a(4 + 3ba)$$

What does this tell us about f with respect to a ?

Some Evaluation

We've recovered

$$\frac{\partial}{\partial a} \left[2a^2 + a^3b \right] = a(4 + 3ba)$$

What does this tell us about f with respect to a ?

Now, let's calculate $\frac{\partial f}{\partial b}$. What does this tell us about f with respect to b ?

Some More Examples

The exact same is true no matter how many terms. For example, solve $\frac{\partial}{\partial k}$ for each k in the following:

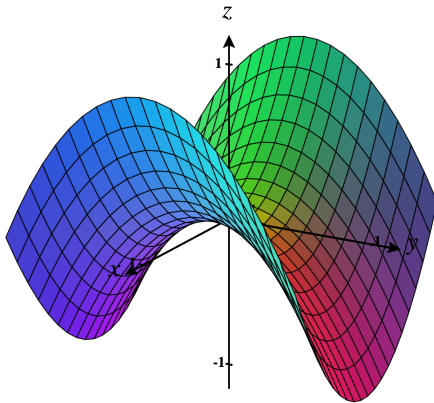
$$2xy^2 + 2x^2y + xy$$

$$3(xy)^2 + \frac{1}{w(x+y)}$$

$$\sqrt{xy} + x^2y^2 + e^{2w}$$

In this way, we can fully characterize each partial derivative

A Grounding Moment



Cross Derivative

We can take a second derivative using this same process, just repeated.
We can also take a **cross derivative**, taking a partial of one variable and then a partial of that result:

$$\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial y} \left[\frac{\partial f}{\partial x} \right]$$

How the derivative of f with respect to x changes with respect to y

Multivariable Optimization

If we are just optimizing a multivariable function w.r.t. one variable, we calculate the first order condition for that variable, and compare to endpoints (this is most common)

Otherwise, we exploit the same logic: what would $\frac{\partial f}{\partial k} \neq 0$ imply for any k ?

Critical Points

Just like single-variable functions, we begin with **critical points** where $\frac{\partial f}{\partial k} = 0$

But, we extend this: it must be true that $\frac{\partial f}{\partial k} = 0 \forall k \dots$

We often get a **system of equations**

Systems of Equations

Given a series of equalities (or inequalities), we must satisfy all of them.
We can use

1. Substitution – isolate for one variable, plug into others
2. Addition/subtraction (messy)
3. Computation

An Example

Take from above:

$$f(x, y) = 2xy^2 + 2x^2y + xy$$

What are our critical points?

An Example

Take from above:

$$f(x, y) = 2xy^2 + 2x^2y + xy$$

What are our critical points?

Solve for partial derivatives:

$$\frac{\partial f}{\partial x} = 2y^2 + 4xy + y = y(2y + 4x + 1)$$

$$\frac{\partial f}{\partial y} = 4xy + 2x^2 + x = x(2x + 4y + 1)$$

When are both of these equal to zero?

An Example

$$y(2y + 4x + 1) = 0 \iff y \in \left\{0, \frac{-4x - 1}{2}\right\}$$

$$x(2x + 4y + 1) = 0 \iff x \in \left\{0, \frac{-4y - 1}{2}\right\}$$

This yields four combinations:

$$(x^*, y^*) \in \left\{ \left(0, 0\right), \left(\frac{-4y - 1}{2}, 0\right), \left(0, \frac{-4x - 1}{2}\right), \left(\frac{-4x - 1}{2}, \frac{-4y - 1}{2}\right) \right\}$$

For our internal points, we have the other variable: $y = x = \frac{-1}{2}$. Then, we solve our system of equations, to recover our critical points

$$(x^*, y^*) \in \left\{ \left(0, 0\right), \left(\frac{-1}{2}, 0\right), \left(0, \frac{-1}{2}\right), \left(\frac{-1}{6}, \frac{-1}{6}\right) \right\}$$

End Points

Both of our partial derivatives are strictly increasing in their variables, so we also examine the points at $(\pm\infty, \pm\infty)$:

$$f(\infty, \infty) = \infty$$

$$f(\infty, -\infty) = \text{undefined}$$

$$f(-\infty, -\infty) = \infty$$

$$f(-\infty, \infty) = \text{underfined}$$

Therefore, maxima are at (∞, ∞) and $(-\infty, -\infty)$

What about minima?

Let's discard our end points here (I didn't think about them when choosing this function), and evaluate every critical point:

$$f(0, 0) =$$

$$f\left(\frac{-1}{2}, 0\right) =$$

$$f\left(0, \frac{-1}{2}\right) =$$

$$f\left(\frac{-1}{6}, \frac{-1}{6}\right) =$$

The minima of f is therefore at the lowest-value critical point here.

Multivariate Integration

Integration of a multivariable function works just like a partial derivative:

1. Identify the variable of integration:

$$\int 2a^2 + a^3b \, da = \text{Integration of } f(a, b) \text{ over } a$$

2. Treat all other variables (i.e., b) as constants, then follow normal derivation rules for a :

$$\begin{aligned} &\Rightarrow \frac{2a^{2+1}}{2+1} + \frac{ba^{3+1}}{3+1} \\ &\Rightarrow \int f(a, b) \, da = \frac{2a^3}{3} + \frac{ab^4}{4} \end{aligned}$$

Recall that the integral of k over a constant m is $k \cdot m$, so $\int f \, da = f \cdot a$ for a term excluding a .

Some More Examples

Solve $\int f \, dk$ for each k in the following:

$$2xy^2 + 2x^2y + xy$$

$$3(xy)^2 + \frac{1}{wxy}$$

$$\sqrt{xy} + x^2y^2 + e^w$$

A Brief Note

Recall our product corollary of integration:

$$\int b \cdot f(x) dx = b \cdot \int f(x) dx$$

With multivariable functions, this holds:

$$\int b \cdot f(x, y) dy = b \cdot \int f(x, y) dy$$

We treat non-integrated variables as constants, so... **isolate whenever possible!** i.e.

$$\int 2yx^2 + 8y^2x dx \implies 2y \cdot \int x^2 + 4yx dx$$

Iterated Integration

Sometimes, we need to integrate over multiple dimensions. For $\mathbb{R}^2$, we call this **double integration**; for $\mathbb{R}^3$, **triple integration**, etc.

$$\int \int f(x, y) \, dx dy$$

Iterated Integration

Sometimes, we need to integrate over multiple dimensions. For $\mathbb{R}^2$, we call this **double integration**; for $\mathbb{R}^3$, **triple integration**, etc.

$$\int \int f(x, y) \, dx dy$$

$$\underbrace{\int \underbrace{\int f(x, y) \, dx}_{\text{Step 1}}}_{\text{Step 2}}$$

Hint: inside $\rightarrow$ out

Iterated Integration (Example)

Let $f(x, y) = \frac{1}{2}x^2y + 2xy^2$. Evaluate

$$\int \int f(x, y) \, dx \, dy$$

Iterated Integration (Example)

Let $f(x, y) = \frac{1}{2}x^2y + 2xy^2$. Evaluate

$$\int \int f(x, y) \, dx \, dy$$

Now, evaluate

$$\int \int f(x, y) \, dy \, dx$$

Iterated Integration (Example)

Let $f(x, y) = \frac{1}{2}x^2y + 2xy^2$. Evaluate

$$\int \int f(x, y) \, dx \, dy$$

Now, evaluate

$$\int \int f(x, y) \, dy \, dx$$

This is called **Fubini's theorem**

For Definite Boundaries

We can calculate a definite iterated integral just using the same steps above (just with FTOC).

Let's evaluate, for the square from $(0, 0)$ to $(2, 2)$, the following:

$$\int \int 2^x + x\sqrt{y} \, dx \, dy$$

Closing Exercise

The following is used by the NWS to calculate wind chill:

$$W(t, v) = 35.74 + 0.6215t - 35.75(v^{0.16}) + 0.4275t(v^{0.16})$$

You can either wear a windbreaker to decrease perceived wind speed, or a jacket to increase perceived temperature, but a lion with your name and phone number on its collar escapes from the zoo² if you wear both. Under what conditions would you want to wear one over the other? (Assume both garments are equivalently weak and only help a little.)

²right next to the preschool